

**YEAR 10 SCIENCE
EARTH AND SPACE SCIENCES
REVISION**

1. Complete the following table.

TERM	DEFINITION
light year	Distance travelled by light in 1 year.
constellation	Pattern formed by stars.
galaxy	Collection of stars held together by gravity.
nebula	Gas cloud.
nova	Small explosion of medium-size star.
protostar	Formation of a star. Fusion starts due to heat + pressure.
red giant	Medium stars swell in size as fuel runs out.
neutron star	Dense star made of neutrons only.
black hole	Very dense + huge gravity. Light can't escape.
black dwarf	Cold matter left after medium star dies.

2. Light travels at 300,000 km/sec. A star is 65.0 light years away.

- (a) How many seconds are there in 65.0 years?
(365 days x 24 hours x 60 mins x 60 secs)

$$2.05 \times 10^9 \text{ s} \quad (\text{used } 365 \text{ days})$$

- (b) Calculate how many kilometres are in 65.0 light years.
(distance = speed x time)

$$v = 3.0 \times 10^5 \text{ km/s}$$

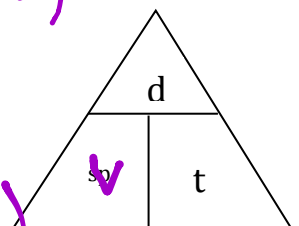
$$d = ?$$

$$t = 2.05 \times 10^9 \text{ s}$$

$$d = v \times t$$

$$= (3.0 \times 10^5)(2.05 \times 10^9)$$

$$= \underline{6.15 \times 10^{14} \text{ km}}$$



- (c) With our current technology, are we likely to be able to travel to this star in the near future? Give a few reasons for your answer - a calculation of time would be very good.

(Assume the fastest rockets can travel at 100,000 km/h - about 28 km/sec.)

YES / ☒ NO (Circle your answer.)

Reasons:

- Rockets are too slow.
- Huge distances to cover.
- Time = 1000's of years!

3. (a) Name the process responsible for producing the energy released from the Sun.

Fusion

- (b) What is the major element found in the Sun?

Hydrogen

- (c) Describe how this element is involved in producing energy.

- Hydrogen changes to helium.
- Some mass is "lost".
- "Lost mass" becomes energy according to $E = mc^2$.

- (d) The Sun loses 5 million tonnes (5.00×10^9 kg) of mass per second. According to Einstein's equation ($E = mc^2$), how much energy (in Joules - J) is released?

$$E: ?$$
$$m = 5.0 \times 10^9 \text{ kg}$$
$$c = 3.0 \times 10^8 \text{ m/s}$$

$$E = mc^2$$
$$= (5.0 \times 10^9) (3.0 \times 10^8)^2$$
$$= \underline{4.5 \times 10^{26} \text{ J}}$$

4. How is the surface temperature of a star related to its colour?

TEMPERATURE	COLOUR
cool	red
moderate	yellow
hot	blue/white

5. (a) About how old is our Sun? 5 billion years

- (b) How much longer will it "live"? 5 billion years

6. Describe how does a nebula become a protostar?

- Gas gathers in the nebula.
- Gravity pulls the material together.
- As the mass increases, heat is generated as the pressure increases.
- Temperature reaches ~ 1 million $^{\circ}\text{C} \Rightarrow$ Fusion starts!

7. What is the major factor that determines the pathway a star takes in its life once it has formed?

- How big/massive it is.

8. (a) Draw a flow diagram (with arrows) to show the stages of our Sun's remaining life as it runs out of fuel.

Sun $\rightarrow$ Red giant $\rightarrow$ Nova $\rightarrow$ White dwarf $\rightarrow$ Black dwarf

- (b) Do similar flow diagrams for the following types of stars.

Huge star $\rightarrow$ Red supergiant $\rightarrow$ Supernova $\rightarrow$ Neutron star

Giant star → ^{Red}supergiant → Supernova → Black hole

- (c) What is the difference between a **nova** and a **supernova**?

Nova: "Small" explosion of medium star.

Supernova: "Large" explosion of massive star.

- (d) What is the difference between a **black dwarf** and a **black hole**?

Black dwarf: Small, cold matter left after medium star dies.

Black hole: Small, super dense with huge gravity; left after a massive star dies.

9. Why do the stars appear to shift slightly in the night sky throughout the year?

- Earth moves $\frac{1}{2}$ way around Sun in 6 months.
- Closer stars move relative to distant stars.

10. How did the observation of the stellar parallax effect change our ideas about the universe?

- Scientists thought all stars were fixed on a sphere.
- Parallax showed they were at different distances from Earth.

11. All of the stars in the constellation Orion appear to be at the same distance from Earth. Explain how observers on Earth know they are not.

- Using spectral analysis and Doppler shift, the stars are at different distances.

12. Why are most stars found on the main sequence of the Hertzsprung-Russell diagram?

- They are stable.
- Spend most of their lives there.

13. Why are there dark lines on the spectrum of light emitted by stars?

- Gases around stars absorb certain frequencies/colours of light.
- Appear as dark lines on a rainbow spectrum.

14. (a) Which colour - red or blue - has the highest frequency?

Blue

(b) What is a red shift? What does it tell us about how a star or galaxy is moving relative to Earth.

- Dark lines appear moved towards the red end of the spectrum.
- Galaxy is moving away, causing wavelength to be longer (towards red).

(c) What did Edwin Hubble find out about the movement of distant galaxies?

- Distant galaxies are red shifted.
- These galaxies are moving fastest.
- Implies the universe is expanding.

15. (a) How old is the universe thought to be?

13.7 billion years

(b) According to the Big Bang Theory, what was present at the beginning of the universe?

- Singularity - very dense point full of energy.

(c) Give a brief description of how the Big Bang Theory describes the formation of the universe.

- Singularity expands, creating space + time.
- Energy becomes matter, according to $E=mc^2$.

(d) The Steady State Theory was in competition to the Big Bang Theory. Briefly describe this theory.

- Universe has always existed and is of uniform density.
- As it expands, new material is created between galaxies.

(e) How did the Steady State Theory explain that the universe was expanding, yet remained the same?

- As it expands, new material is created between galaxies.
- Average density of the universe remains constant.

(f) What evidence put an end to the Steady State Theory?

- Cosmic background radiation, which indicated the presence of "a big expansion".

(g) Describe 3 major pieces of evidence that supported the Big Bang Theory.

- Red shift of galaxies moving away from each other.
- Cosmic background radiation from the energy present at the "big expansion".
- Amount of hydrogen + helium in the universe.